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$$E = -\frac{e^2}{2an^2} \quad D = n^2$$

精细结构: 由一阶相对论效应和自旋-轨道耦合产生的谱线分裂为精细结构

起~: 用高分辨率光谱学方法研究原子谱时, 可发现原子谱线由多条线组成, 呈现出非常精细的结构

(5) 氢原: $E = -\frac{e^2}{2an^2} \quad a = 0.07 \text{ nm}$
 $E = -\frac{e^2}{2(0.07 \text{ nm})n^2}$

二. 1. $x=a$ 处: $\psi'(a^+) - \psi'(a^-) = \frac{2mA}{\hbar} \psi(a) \quad \psi(a^+) = \psi(a^-)$
 $x=-a$ 处: $\psi'(-a^+) - \psi'(-a^-) = \frac{2mA}{\hbar} \psi(-a) \quad \psi(-a^+) = \psi(-a^-)$

若取 $x=a$ 处, 有 ~~$\sinh k_2 x$~~ 保留下.

$$\begin{cases} A \sinh k_1 a = B e^{-k_2 a} \\ k_1 B e^{-k_2 a} + k_2 \sinh k_2 a = \frac{2mA}{\hbar} B e^{-k_2 a} \end{cases}$$

$$\Rightarrow B = \frac{\hbar}{2mA} e^{2k_2 a}$$

同理对于 $x=a$ 处

$$C = \frac{\hbar}{2mA} e^{2k_2 a}$$

(3) 满足?

(4) 存在解, 需要满足在 $x=0$ 处, 解连续 (左, 右) ($x \neq 0$)

up: $k_1 < 1 + k_2$

$x=0$ 时 同样存在解.

$$H = AS_2 + BS_x^2$$

up $S^2 = S(S+1)\hbar^2 = 2\hbar^2 \Rightarrow S_x^2 + S_y^2 + S_z^2 \Rightarrow S_x^2 = \frac{2}{3}\hbar^2$

$$1/4 AS_2 + \frac{2}{3} BS_x^2$$

则 S_x^2 与 S_y^2 与 S_z^2 同时为 0

$$E_1 = \frac{1}{4}A\hbar^2 + \frac{2}{3}B\hbar^2, \quad \psi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad E_2 = \frac{1}{4}A\hbar^2 + \frac{2}{3}B\hbar^2, \quad \psi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

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$$H = \frac{1}{2} A \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + B \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a+b & b \\ b & -a+b \end{pmatrix}$$

其中 $a = Ah$, $b = B\frac{h^2}{2}$

$$H\psi = E\psi \Rightarrow \begin{vmatrix} a+b-E & b \\ b & -a+b-E \end{vmatrix} = 0$$

$$\Rightarrow E_1 = 2b, E_2 = b + \sqrt{a^2 + b^2}, E_3 = b - \sqrt{a^2 + b^2}$$

对于 E_1, E_2, E_3 其对应的本征态分别为: 不妨设 $E_1 = 2m, E_2 = m+n, E_3 = m-n$

$$|4_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |4_2\rangle = \frac{1}{\sqrt{(m-a)^2 + m^2}} \begin{pmatrix} m \\ m \\ m-a \end{pmatrix}, |4_3\rangle = \frac{1}{\sqrt{(m+a)^2 + m^2}} \begin{pmatrix} m \\ m \\ -m-a \end{pmatrix}$$

同时 $|4_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$$c_1 = \langle 4_1 | 4_1 \rangle = 0$$

$$c_2 = \langle 4_2 | 4_1 \rangle = \frac{m}{\sqrt{(m-a)^2 + m^2}}$$

$$c_3 = \langle 4_3 | 4_1 \rangle = \frac{m}{\sqrt{(m+a)^2 + m^2}}$$

$$|4\rangle = \sum_i c_i |4_i\rangle e^{-iE_i t/\hbar} = c_2 |4_2\rangle e^{-iE_2 t/\hbar} + c_3 |4_3\rangle e^{-iE_3 t/\hbar}$$

$$= \frac{1}{\sqrt{(m-a)^2 + m^2}} \begin{pmatrix} m \\ m \\ m-a \end{pmatrix} e^{-iE_2 t/\hbar} + \frac{1}{\sqrt{(m+a)^2 + m^2}} \begin{pmatrix} m \\ m \\ -m-a \end{pmatrix} e^{-iE_3 t/\hbar}$$

其中 $m = b, B\frac{h^2}{2}, n = \sqrt{a^2 + b^2} = \sqrt{A^2 h^2 + \frac{B^2 h^4}{4}}$

$$E_2 = B\frac{h^2}{2} + \sqrt{A^2 h^2 + \frac{B^2 h^4}{4}}, E_3 = B\frac{h^2}{2} - \sqrt{A^2 h^2 + \frac{B^2 h^4}{4}}$$

四. 4: $A e^{-\frac{\alpha^2 x^2}{2}}$

1. $\int_{-\infty}^{\infty} |4|^2 dx = \int_{-\infty}^{\infty} A^2 e^{-\alpha^2 x^2} dx = A^2 \frac{\sqrt{\pi}}{\alpha} = 1 \Rightarrow A = \sqrt{\frac{\alpha}{\pi}}$

2. $4_0 = \int_{-\infty}^{\infty} e^{-\alpha^2 x^2} dx$

$$\langle x \rangle = \langle 4_0 | x | 4_0 \rangle = 0, \langle p \rangle = \langle 4_0 | p | 4_0 \rangle = 0$$

$$\langle x^2 \rangle = \langle 4_0 | x^2 | 4_0 \rangle = \frac{\hbar^2}{2m} \left((2x+1) \delta_{00} + \sqrt{2} \delta_{02} \right) = \frac{\hbar^2}{2m}$$

$$\langle p^2 \rangle = \langle \psi_0 | p^2 | \psi_0 \rangle = \langle p \psi_0 | p \psi_0 \rangle$$

$$p \psi_0 = \frac{\hbar}{i} \frac{d}{dx} \left(\frac{1}{\sqrt{2a}} e^{-\frac{x}{2a}} \right) = -\frac{\hbar}{2a} e^{-\frac{x}{2a}}$$

$$\langle p^2 \rangle = \frac{\hbar^2}{4a^2} \int_{-\infty}^{\infty} e^{-x/a} dx = \frac{\hbar^2}{4a^2} \cdot 2a = \frac{\hbar^2}{2a^2}$$

$$4) \quad \psi_0 = \frac{1}{\sqrt{2a}} e^{-\frac{x}{2a}}$$

$$E_n = (n + \frac{1}{2}) \hbar \omega, \quad \psi_n = \frac{1}{\sqrt{n!}} (a^\dagger)^n \psi_0$$

$$4) \quad \Delta x \cdot \Delta p^2 = \frac{\hbar}{2m\omega} \cdot \frac{\hbar^2}{4} m\omega = \frac{\hbar^2}{8} \quad (\text{基态})$$

$$\text{对于 } n \neq 0 \quad \Delta x \cdot \Delta p^2 = (n + \frac{1}{2})^2 \cdot \hbar^2$$

$$\text{例} \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar m \omega}} (-i\hat{p} + m\omega \hat{x}) \quad \hat{a} = \frac{1}{\sqrt{2\hbar m \omega}} (i\hat{p} + m\omega \hat{x})$$

$$\text{又, } H = \begin{pmatrix} E_1 & a \\ & \bar{a} \end{pmatrix} \quad |4_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |4_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$E_1^{(1)} = H_{11}^{(1)} = a$$

$$E_2^{(1)} = H_{22}^{(1)} = \bar{a}$$

$$E_1^{(2)} = \sum_{m \neq 1} \frac{|H_{1m}^{(1)}|^2}{E_1 - E_m} = \frac{\beta_2^2}{E_1 - E_2}$$

$$E_2^{(2)} = \sum_{m \neq 2} \frac{|H_{2m}^{(1)}|^2}{E_2 - E_m} = \frac{\beta_1^2}{E_2 - E_1}$$

$$\text{修正 } |4_1\rangle = \frac{\beta_2}{E_1 - E_2} |4_2\rangle \quad |4_2\rangle = \frac{\beta_1}{E_2 - E_1} |4_1\rangle$$

$$4) \quad H = \begin{pmatrix} E_1 & \\ & E_2 \end{pmatrix} \quad |4_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |4_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad H = \begin{pmatrix} a & \\ & \bar{a} \end{pmatrix}$$

$$\text{双能级系统 } \psi = c_1 |4_1\rangle + c_2 |4_2\rangle$$

$$\begin{pmatrix} H_{11} - E & H_{12} \\ H_{21} & H_{22} - E \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0$$

$$H_{11} = \langle 4_1 | H | 4_1 \rangle = a, \quad H_{12} = 0, \quad H_{21} = 0, \quad H_{22} = \bar{a}$$

$$E = a, \quad \text{解并未消除, 无法确定 } |4\rangle$$